

Mechanical simulation analysis of the failure risks and influence factor of foldable module

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Abstract

In order to ensure the foldability of the foldable screen, it has to maintain a low fold stress through optimization of the foldable modulus stacking structure and materials. Nonlinear mechanical simulation can be an effective tool to provide insights into folding behavior. Previous simulation only focused on analyzing the stress of foldable module in the folded state. In this work, the stress of the module was analyzed not only in the folded state but also during the unfolding process by simulation. Different folding programs considering the creep of the adhesive layer in a bending state were simulated. From these results, we confirm that the load history may have a significant impact on the stress of the foldable module. Long term retention in a folded state significantly increases the risk of failure during the unfolding process. It is necessary to consider the influence of the folded state maintenance process when designing experimental schemes to test the folding ability of foldable module. Furthermore, we confirm that these risks are related to creep&recovery performance of the adhesive layer, and a material with low creep or high instantaneous creep recovery rate is required to increase the foldability.

Author Keywords

Foldable display; Mechanical Simulation; Foldability; Optical Clear Adhesive; Creep&Recovery.

1. Introduction

The foldable display market has been significantly growing since Samsung released the first foldable mobile phone Galaxy fold in 2019. The water-drop shape inward foldable display products are a mainstream type due to its compactness than wedge shape, but at the same time, the water-drop shape require a much more complicated hinge design. In the past two years, with the rapid development of hinge technology, foldable mobiles have become thinner and lighter. For example, the thickness of honor MagicV2 has been reduced to below 10mm when folded. On the one hand, these extreme designs improve the user's experience, and on the other hand, these designs put forward higher mechanical reliability requirements for foldable module.

In order to ensure the foldability of the foldable screen, it has to maintain a low fold stress through optimization of the foldable module stackup and materials. The stress and strain distribution generated in each layer can be analyzed using finite element method [1]. In recent years, this lead to the development of various simulation models to predict the reliability of multilayered structures, to gain structural understanding, and to improve design.

Existing simulation only focused on analyzing the stress of foldable module in the folded state [2,3]. Based on these analysis, the adhesive failure that occurs in the outward folding area of the water-drop shape structure can be well understood. Adhesive failure is usually related to micro defects such as cavities [4]. The hydrostatic tension stress plays an important driving role in the growth of voids, therefore, the debonding index based on pressure instead of the maximum stress parameter can be used to

effectively evaluate the risk of adhesive failure [5]. Under the guidance of these simulation models, an optimized folding module stackup can even withstand up to 1 million bending cycles under laboratory testing conditions. Despite this, foldable screens remain susceptible to failure during operation, especially when the folding radius is very small. Some failure phenomena, such as OCA (Optical clear adhesive) bubbles and black lines (figure1) occurring at the bending center still have no reasonable explanation and is still a need for more in-depth study. Under laboratory testing conditions, the display module is usually folded and opened continuously at a constant speed. However, in actual consumer usage scenarios, foldable phones undergo more complex folding conditions.

In this work, the stress of the module was analyzed not only in the folded state but also during the unfolding process by simulation. Different folding programs considering the creep of the adhesive layer in a bending state and the unfolding speed of the foldable module were simulated. From these results, we confirm that the load history has a significant impact on the stress of the module. Long term retention in a folded state significantly increases the risk of failure during the unfolding process. Furthermore, we attribute these risks to the viscoelasticity of the adhesive layer, and a material with low creep or high instantaneous creep recovery rate is required to increase the foldability.



Figure 1. Failure phenomena of foldable phone.

2. Simulation

The folding and unfolding procedure of foldable module is simulated by using nonlinear finite element software to analyze the bending stress. In order to obtain more accurate results, a visco-hyperelastic OCA constitutive is adopted. It is worth noting that selecting a constitutive model capable of accurately depicting the hyperelastic and viscoelastic characteristics of OCA during large deformation in simulation is crucial.

A typical foldable stack structure is considered in this work. The typical film stack includes: cover film, AMOLED panel, the back film, the metal support sheet. The AMOLED panel itself also consists of multiple components including a flexible substrate, thin film transducer (TFT), thin film encapsulation (TFE), the

touch sensor layer. A schematic diagram of stack structure is shown in figure2a. A water-drop shape structure with radius of $\sim 1.5\text{mm}$ as shown in figure2b is considered in this work.

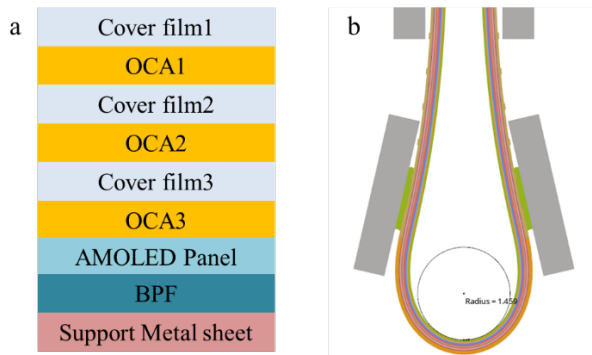


Figure 2. A schematic diagram of stack structure (left) and water-drop shape (right) used in this work.

To investigate the influence of load history on fold stress, three sets of folding programs were considered. In case A, the folding process takes 1 s and then immediately unfolded within 1 second. In case B, the folding process takes 1 s, and the display remains in the folded state for a predetermined duration of 600 s, after that, the display is opened within 1 second. In case C, the module was folded within 1 second and after being folded for 600 s, it is opened within 0.1 second. Detailed information is presented in table1.

Table 1. Three sets of folding programs

	Case A	Case B	Case C
Folding	1s	1s	1s
Keeping	/	600s	600s
Unfolding	1s	1s	0.1s

Unless otherwise specified, we use the 2D plane strain model to represent the stack-up due to the thin film nature of the foldable module. In order to study the influence of pattern of supporting metal sheet, a detailed 3D structure is also simulated.

3. Result and discussion

Load history effect:

Figure 3a and figure 3b show the variation of OCA3 pressure stress with folding angle and unfolding angle respectively. A debonding risk index based on pressure is used to evaluate the debonding risk. During the folding process, as the folding angle increases, the OCA pressure stress increases. When the module is fully folded, the OCA pressure stress reaches its maximum value. The maximum stress of OCA occurs near the outward folding area of the water drop structure (figure 4a), the most common debonding failure of OCA is found at this area in typical dynamic bending tests. Because the pressure stress of OCA is below, based on our experience, the risk of OCA failure in this area is low under current design. For different folding programs, the pressure stress of OCA shows different trends with increasing unfolding angle. For case A, the OCA stress remains at very low level throughout the entire unfolding process. However, for case B, it is found that

keeping the folded state for a period of time before unfolding significantly increases the OCA pressure stress during the unfolding process. It should be noted that the peak stress of OCA during the unfolding process is significantly higher than that in the folded state. Furthermore, the location where the maximum pressure stress of OCA occurs also shifts from the outward folding area to the bending center. Simulation predicts that OCA failure may also occur in the center of the bending region corresponding to the OCA bubble location shown in figure1b. Increasing the unfolding speed will further increase OCA stress as can be seen from the result for Case C.

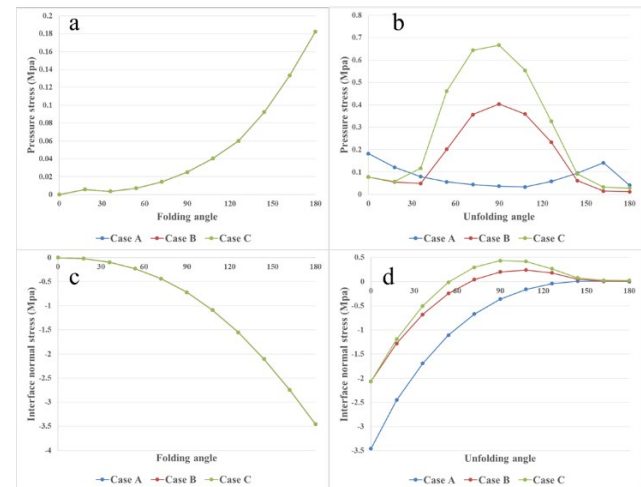


Figure 3. (a) The variation of OCA3 pressure stress during folding process. (b) The variation of OCA3 pressure stress during unfolding process. (c) The variation of interface normal stress during folding process. (d) The variation of interface normal stress during unfolding process.

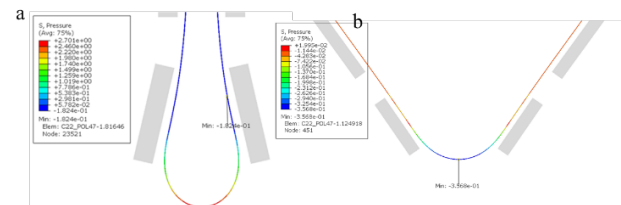


Figure 4. (a) The pressure distribution shows that the maximum stress occurs in the outward folding area. (b) The pressure distribution shows that the maximum stress occurs in the bending center area.

The black line shown in figure1 usually corresponds to interlayer delamination of the panel film which is related to interface stress. Therefore, we also analyzed the interface normal stress of weak interfaces in the panel. We focused our attention on the central area of the bending zone. Figure 3c and figure 3d show the variation of interface normal stress with folding angle and unfolding angle respectively. During the entire folding process, the normal stress at the interface is compressive stress, which means that there is no risk of delamination failure of the panel in these areas during folding process. For typical dynamic bending test, the foldable module was opened immediately after been folded as case A. the results indicate that the normal stress at the interface remains compressive through the unfolding process.

However, when keeping the folded state for a period of time before unfolding as case B and case C, the result shows that the interface stress will transition from compressive stress to tensile stress during unfolding process. Furthermore, fast unfolding speed corresponds to greater interface normal stress.

OCA viscoelasticity effect:

As discussed in the previous section, the failure risk during the unfolding process stems from the creep of the adhesive layer, which cannot promptly restored during the unfolding process. Therefore, it can be inferred that a material with low creep or high instantaneous creep recovery rate help improve foldability of foldable module. In order to investigate the influence of the creep&recovery performance of OCA on bending stress, three kinds of OCA with different creep&recovery performance but similar modulus were compared and analyzed in our simulation. The key parameters are presented in table2. OCA-A has the maximum creep deformation and the minimum instantaneous creep recovery rate, OCA-C has the minimum creep deformation and the maximum instantaneous creep recovery rate. The instantaneous creep recovery rate refers to the proportion of the deformation recovery relative to the maximum deformation at the moment when the load is completely unloaded to zero.

Table 2. Creep&recovery performance of different OCA considered in this work

	OCA-A	OCA-B	OCA-C
Modulus	40kpa	40kpa	40kpa
Creep strain	620%	290%	130%
creep recovery rate (instantaneous)	30%	51.7%	76.9%

Figure 6a shows the variation of OCA pressure stress with unfolding angle during unfolding process. Figure 6b shows the variation of interface normal stress with unfolding angle during unfolding process. As the creep deformation of OCA decreases or the instantaneous creep recovery rate increases, both OCA pressure stress and interface normal stress decrease. When OCA-B and OCA-C are used to replace OCA-A, the pressure stress of OCA during unfolding decreases by 25.3% and 71.6% respectively, interface the normal stress decreases by 27.9% and 95% respectively. Therefore, our results confirm that controlling the creep and recovery performance is crucial for improving the foldability of foldable module.

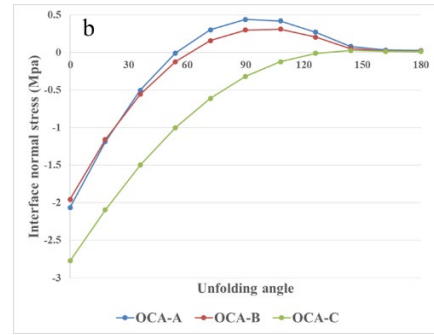
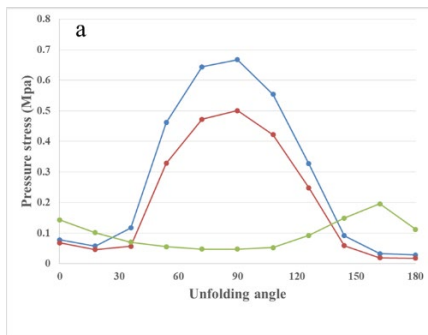


Figure 6. (a) The variation of OCA3 pressure stress during unfolding process. (b) The variation of interface normal stress during unfolding process.

4. Conclusion

In this work, the bending stress of foldable module was analyzed by nonlinear mechanical simulation. The stress of the module was analyzed not only in the folded state but also during the unfolding process. A visco-hyperelastic OCA constitutive is adopted, which is crucial for obtaining results.

From these results, we confirm that the load history may have a significant impact on the stress of the foldable module. Long term retention in a folded state significantly increases the risk of failure during the unfolding process. It is necessary to consider the influence of the folded state maintenance process when designing experimental schemes to test the folding ability of foldable module. Furthermore, we confirm that the main key factor are creep&recovery performance of OCA to improve foldability, and a material with low creep or high instantaneous creep recovery rate is required.

5. References

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